

Nisanur Camuzcu

Education

- 2024–Present **Doctor of Philosophy (Ph.D.)**, *Rochester Institute of Technology, Kate Gleason College of Engineering*, Rochester, NY, | Expected Graduation: **May 2028**
Electrical & Computer Engineering | Advisor: Prof. Alireza Vahid | **GPA: 4.00/4.00**
- 2022–2024 **Master of Science (M.Sc.)**, *Politecnico di Milano*, Milan, Italy
Telecommunication Engineering | **CGPA: 93/110** [\[Thesis\]](#)
- 2017–2022 **Bachelor of Science (B.Sc.)**, *Bilkent University*, Ankara, Turkey
Electrical and Electronics Engineering | **CGPA: 3.10/4.00**

Research Interests

Wireless Communications | LEO Satellite Navigation and PNT | Localization and Sensing | Communication Networks

Publications

- 2026 **Secure Pseudonymity with DSSS Watermarking for LEO Satellites**
N. Camuzcu, A. Vahid
IEEE Military Communications Conference (MILCOM 2026), Accepted. [\[Link\]](#)
- 2026 **Starlink Beacons for Passive LEO-Aided 9D Navigation**
N. Camuzcu, T. M. Hoang, A. Vahid
IEEE 104th Vehicular Technology Conference (VTC2026-Fall), Accepted. [\[Link\]](#)
- 2026 **Traffic-aware network reconfiguration for energy-efficient IPoWDM networks with ZR/ZR+ plug-gable optics**
Q. Zhang, N. Camuzcu, X. Guo, J. Xiong, R. Wang, T. Luo, M. Ibrahimi, M. Tornatore
ITU Journal on Future and Evolving Technologies. [\[Link\]](#)
- 2024 **Minimizing Power Consumption in IPoWDM Networks Using ZR/ZR+ with Network Re-Configuration**
N. Camuzcu, Q. Zhang, M. Ibrahimi, R. Wang, L. Zhang, M. Tornatore
Asia Communications and Photonics Conference (ACP 2024) – Best Paper Award. [\[Link\]](#)
- 2022 **Relative Localization for Swarm Aerial Vehicles**
A. Aytekin, F. Bagci, M. Bozdog, N. Camuzcu, A. Duru, M. S. Elma, A. Maghsoudi, B. Barshan
IEEE 30th Signal Processing and Communications Applications Conference (SIU 2022). [\[Link\]](#)

Research Experience

- Mar. 2026–Present **Agentic AI for Robust Passive LEO-Aided Navigation**
Advisor: Prof. Alireza Vahid
- Developing an agentic supervisory framework for adaptive trust orchestration across GNSS, LEO satellite, and inertial measurements under jamming, spoofing, and changing satellite availability.
 - Designing belief-state reasoning, measurement gating, recovery logic, and online bias mitigation for robust 9D navigation, with comparison against non-agentic baselines and ablations.
- 2026–Present **DSSS Watermarking for Passive LEO Signal Attribution**
Advisor: Prof. Alireza Vahid
- Developing a low-power direct-sequence spread-spectrum (DSSS) watermarking framework for passive identification and authentication of LEO satellite signals without requiring payload demodulation.
 - Designed pseudonymous satellite identification and keyed authentication mechanisms for reliable source attribution under weak-signal and multi-satellite conditions.
 - Resulted in an invention disclosure submitted for provisional patent protection.
- Jul. 2025–Feb. 2026 **Passive LEO-Aided 9D Navigation Using Starlink Beacons**
Advisors: Prof. Alireza Vahid, Tiej M. Hoang
- Developed and experimentally validated a passive GPS/LEO/IMU navigation architecture using Starlink beacon Doppler dynamics without satellite synchronization or active transmission.
 - Captured and processed real satellite signals using low-cost RF hardware and demonstrated LEO-assisted navigation under degraded GPS conditions.

- Sep. **RIS-Assisted ToF Localization in mmWave WiFi**
2024–Aug. Advisors: Prof. Alireza Vahid, Tiep M. Hoang | *Under Review*
2025
 - Developed a ToF-based mmWave WiFi localization framework using RIS-assisted propagation, beam selection, and statistical timing-bias compensation.
 - Evaluated localization under LoS/NLoS propagation, mobility, RIS configurations, and Gaussian-process-based clock-bias correction.

Teaching Experience

- Spring 2026 **Graduate Teaching Assistant – EEEE-484 Communication Systems**, *Rochester Institute of Technology*, Rochester, NY
Assisted with analog and digital communication topics including modulation, spectral analysis, noise, and detection through laboratory sessions, office hours, and student support.
- Fall 2025 **Graduate Course Content Developer – EEEE-789-Spectrum Sharing & Management**, *Rochester Institute of Technology*, Rochester, NY
Developed graduate-level instructional material on spectrum sharing and coexistence in wireless systems, covering spectrum access models, interference management, sensing and coordination mechanisms, and emerging spectrum-sharing applications. Contributed to lecture content, technical examples, and course organization.

Industry Experience

- Nov.2021– **ASELSAN**, *Candidate Engineer – Waveform Design*, Ankara, Turkey
Jun.2022 Investigated error-detection processes and ARQ/HARQ mechanisms for 5G communication systems.
- Summer 2020 **Türk Telekom**, *Network Analysis Intern*, Ankara, Turkey
Analyzed telecommunication services and 3G/4G network quality using Network Performance Monitoring (NPM) tools.
- Summer 2019 **Bilkent Acoustic and Underwater Technologies Research Center (BASTA)**, *Circuit Design Intern*, Ankara, Turkey
Designed, simulated, and implemented a shielded filter circuit for underwater signal transmission and reception.

M.Sc. Projects

- 2023–2024 **Scalable Microservice Architecture for Optical Service Creation and Monitoring**, *Politecnico di Milano / SM Optics*
Developed a containerized optical-network control and monitoring architecture using Django, PostgreSQL, InfluxDB, NATS, Go, Nginx, and Docker. [\[GitHub\]](#)
- 2023 **Spatial Speech Separation**, *Politecnico di Milano / Huawei*
Implemented ToA-based spatial signal acquisition and LMS-based speech separation for a moving source and microphone array in MATLAB.
- 2023 **Polar Codes**, *Politecnico di Milano*
Implemented and evaluated Successive Cancellation (SC) and Successive Cancellation List (SCL) decoding algorithms in MATLAB.
- 2023 **Routing Strategies for Elastic Optical Networks**, *Politecnico di Milano / Nokia*
Implemented and evaluated routing, modulation, and spectrum-assignment strategies in Python, analyzing blocking probability and network efficiency. [\[GitHub\]](#)

Technical Skills

- Programming Python, MATLAB, C/C++, VHDL, Assembly
Tools Linux, Git, Docker, LTSpice, LabVIEW, $\LaTeX$
Wireless / SDR signal processing, GNSS/LEO measurements, statistical estimation, Kalman filtering, mmWave/RIS
Signal simulation
Processing

Awards & Honors

- 2024 **Best Paper Award**, Asia Communications and Photonics Conference (ACP)
2017–2022 Comprehensive **Scholarship**, Bilkent University
2017 Ranked **76th** in the Turkish University Entrance Exam (LYS) among approximately 2 million students